Python - Day 7

Mapping data type:

{{[1,2,3]}} →

→ key → value

→ containing Data Structure → links → key → value

key ↓ Hashable

(

value ↓ Data type

string

int

float

→ {[1,2,3]} → hashab/unhashable

→ Hashable → Immutable

→ Un Hashable → mutable

→ Set only hashable value are stored → x error

→ key → value → mapping data

key ↓ Hashable → ( ), String, int, float, for

([ ] → unhashable,

value → Data type → Data

Dictionary :-

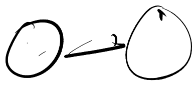
What is existance of Dictionary?

Students :-

{
'Guru' : 23,
'gop' : 22
}

~~name~~ = ["Guru", 'Jaga', "pda', 'gop']

~~age~~ = [23 22 2c 2]



hay

ordered → 1st

→ key-value pairs to store → index

→ collection of ordered data type

→ Mutable and does not allow Duplicity

→ 3.7 → ordered   before 3.7 → unordered.

a hash(key) → number → 1

immutable

Creation of Dict:

→ { }

Hash value

str Hashable

→ Data type ('guru')

str

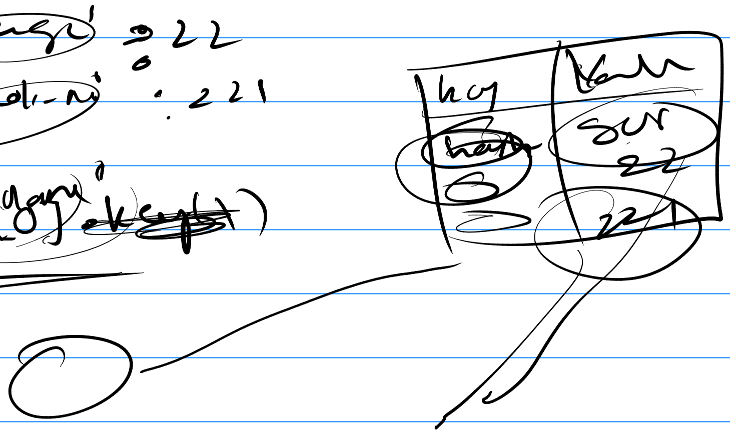
a = {

'Name' : 'guru'

~~'age'~~ : 22

'3Roll_no' : 221

print(a[~~0~~] ~~key~~)



print(a["name"])

print(hash(a["name"]))

→ print(a["name"]) → guru

hash → 1006 → 100 → guru

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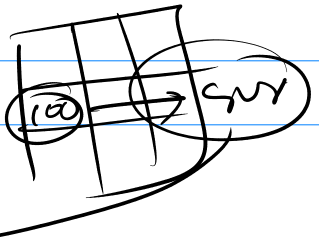
guru



→ { } →

{ → hasble

"key" : "val" → Both Hasble | unshashble

}

Ditctnl items:- key-value

→ order, chagablecand dont alow duplicty

a = {
'age' = 22
~~"age"~~ = 23
} 'age': 24

print(a['age']) #24

Dict length

Using length fun

a = { "nam" : "sur"
'age' : 22
}

1

1+1

print(len(a))

#2

{}

key = value

→ dict() → constructor

→ dict(keyword arguments)

```
a = dict("name" = "guru", "age" = 22)
print(a)        = {"name": "guru", "age": 22}
```

Access items in Dict:

```
a = {"name": "guru", "age": 21}
print(a["name"]) # guru
```

→ get().

```
print(a.get("name")) # guru
```

key:

obj

→ key() → it will return the keys obj of Dictionary

→

```
a = {"name": "guru", age = 22}
b = a.keys()
print(b) #      dict_keys(["name", "age"])
```

↳ view obj

→

```
a = {"name": "guru", "age": 21, "Roll_No": 221}
b = a.key()
a["Roll_No"] = 221
print(b) → ~~{~~dict_key(["name", "age", "Roll_No"])
```

values() :-

→ obj return → only values in Dict

```
a = {"name": "guru", 'age': 21}
a.["Roll_no"] = 221
print(a.values())
```

dict_values(["guru", 21, 221])

items() :-

→ obj return → both key value pairs → Tuples
items → ( ) ← immutable

[

```
a = {"name": "guru", 'age': 21}
a["Roll_no"] = 221
print(a.items())
```

dict_items([('name', 'guru'), ('age', 21), ( )])

( )

```
a = {(1,2,3): "guru", 'name': 'parvan')
```

a.keys() → dict_keys([(1,2,3), 'name'])

(( ), ( ))

→ Dict → Real time ✓ → Dict ✓, list, tuple

JSON → { key → Hashable

}